

Geometry
Unit 12
Lesson 6 Homework

Name **Answer Key** _____
Date _____
Period _____

For questions 1 through 3, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.

$$CD = 3x + 8 \text{ and } DB = 5x - 16.$$

1. What is the value of x ?

$$3x + 8 = 5x - 16$$

$$24 = 2x$$

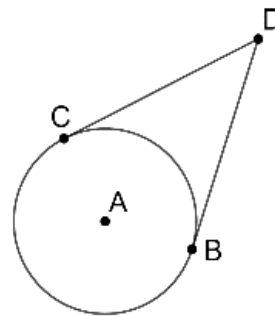
$$12 = x$$

2. What is the length of $\overline{CD}$?

$$3(12) + 8 = 44$$

3. What is the length of $\overline{DB}$?

$$5(12) - 16 = 44$$



For questions 4 through 7, Circle W is shown. Chords $\overline{CB}$ and $\overline{DE}$ intersect at point A . $CA = 3x + 5$, $AB = 3$, $DA = 4$, and $AE = 2x + 4$.

4. Write an equation you can use to solve for x .

$$3(3x + 5) = 4(2x + 4)$$

5. Solve the equation you wrote in question 4 for x .

$$9x + 15 = 8x + 16$$

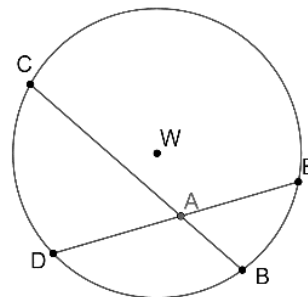
$$x = 1$$

6. What is the length of $\overline{CB}$?

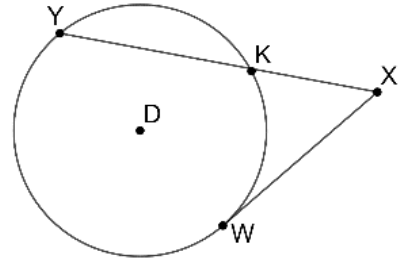
$$3(1) + 5 + 3 = 11$$

7. What is the length of $\overline{DE}$?

$$2(1) + 4 + 4 = 10$$



For questions 8 through 10, Circle D is shown. $\overline{XY}$ is a secant to the circle, and intersects tangent $\overline{XW}$ at point X . The length of $\overline{XW} = 6$, $\overline{KX} = 2$, and $\overline{YX} = 3x + 1$.



8. Write an equation that can be used to find the value of x .

$$2(3x + 1) = 6^2$$

9. Solve the equation you wrote in question 8 to find the value of x .

$$6x + 2 = 36$$

$$6x = 34$$

$$x = 5.\bar{6}$$

10. What is the length of $\overline{YX}$?

$$3(5.\bar{6}) + 1 = 18$$